



Note on Pricing

A price is an expression of value. The value rests in the usefulness and quality of the product itself, in the image that is conveyed through advertising and promotion, in the availability of the product through wholesale and retail distribution systems, and in the service that goes with it. A price is the seller's estimate of what all of this is worth to potential buyers, recognizing the other options buyers will have for filling the need the product is intended to satisfy. To the extent that the product or service finds markets and is profitable at given price levels, it provides a viable economic base for building and maintaining a business.

In the competitive milieu pricing is game playing. The struggle for market share focuses critically on price. Pricing strategies of competing firms, therefore, are highly interdependent. The price one competitor sets is a function not only of what the market will pay but also of what other firms charge. Prices set by individual firms respond to those of competitors; they also are intended often to influence competitors' pricing behavior. Pricing is an art, a game played for high stakes; and for marketing strategists, it is the "moment of truth." All of marketing comes to focus in the pricing decision.

This note deals primarily with pricing strategy. It explores the considerations that come to bear on pricing decisions, and it examines the pricing process. A section at the end of the note delineates certain pricing conventions, or forms, that are commonly used in the pricing of consumer and industrial goods and services. For most of us who see pricing from the buying end of transactions, an examination of pricing strategy from the marketer's perspective should be both fascinating and revealing.

Pricing Strategy

A way to think about making a pricing decision is that price should be set somewhere between what the product costs to make and sell and its value to the customer. If price exceeds the perceived value of the product to potential purchasers, it has no market. If the price is below what the product costs to produce, the business cannot survive for very long. Where a price should be set between cost and customer value is a strategic decision. The factors that influence this decision are competitors' product/price strategies, governmentally imposed constraints, and the seller's and the buyer's sense of what's fair. Finally, the most important determinant of price is the marketer's objectives—what is the firm trying to do?

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We examine each of these six areas in the discussion of pricing strategy that follows and conclude with a treatment of pricing as a dynamic process.

Product Costs

Costs may be classified as variable (or out-of-pocket), fixed, and semifixed. An airline considers the annual depreciation on an aircraft as a fixed cost. Taking the plane off the ground to fly from one city to another incurs certain semifixed costs (e.g., fuel, the compensation of flight personnel, and airport take-off and landing fees). These costs are approximately the same for any given flight whether the plane is empty, half-loaded, or completely full of passengers.

The variable costs of the flight would include primarily the costs of food and beverage. They vary directly with the number of passengers.

In a manufacturing plant making glass bottles, the fixed costs include the depreciation on plant and capital equipment such as furnaces and forklift trucks. The variable costs are what the glass manufacturer pays for energy to fuel the furnaces, sand and limestone to make the glass, and labor. The latter are a direct function of the level of output.

If fixed and semifixed costs make up a large portion of total costs, as in the airline example, pricing to get maximum capacity utilization is crucial. Until the seller covers fixed costs, money is lost. After fixed costs are covered each incremental sale contributes proportionally large amounts to profits.

If variable costs are a relatively high percentage of total costs, as they might be for the glassmaker, pricing to maximize unit contribution (i.e., the difference between unit variable cost and price) on each bottle produced will be critical to profitability. Under these cost conditions, the manufacturer would naturally work to maximize unit prices and to reduce variable costs.

The objective of the airline's pricing strategy will be to generate enough total revenue to cover its fixed costs and above that to get maximum capacity utilization to make profits. The glass company will price to cover its high variable costs per unit and get enough contribution to amortize fixed costs and make a profit.

Under certain conditions companies may elect to price at less than full cost. In conditions of capacity underutilization, for instance, firms with high fixed costs may take business at prices that cover variable costs and make some incremental contribution to fixed costs (or overheads). The hope is to get through bad times, keep the plants running, and hold some critical nucleus of managers, skilled technicians, and labor.

Pricing temporarily at less than full cost may also be used as a strategy to get a particularly large order. The expectation is that by taking the business, the firm may be able to reduce its unit costs and/or later raise its prices so as to make a profit on subsequent orders. Taking business below cost with the hope of offsetting near-term losses with longer-term profits may be a risky tactic, since there is no assurance that the losses can be made up.

Pricing near or below cost may also be done to gain a large market share, a strategy called penetration pricing (on which more will be said later). Generally pricing low to preempt market share is predicated on the assumption that unit costs will come down significantly as volume increases. This may happen through gaining manufacturing experience. In fact, in many businesses a so-called experience (or learning) curve is used to calculate what the effect will be of volume growth on unit costs.

To a large extent, learning curve experience reduces the variable cost component of unit costs. Labor gains in efficiency and purchases of materials and parts in larger volumes all result in lower prices, and manufacturing process improvements produce cost savings.

Indeed, the fixed-cost component of unit costs may also come down with volume increases. Larger plants may be more cost efficient. Large-scale selling and advertising programs may also be more cost efficient. If product sales are particularly sensitive to heavy advertising, or the product requires widespread distribution or extensive field service support, fixed marketing expenditures for these purposes must usually be at a high level.

These so-called *scale economies* come in certain cost categories depending on the product, the processes used to manufacture it, and the level of marketing spending required to be competitive. If significant scale economies are achievable, some competitors may be willing to price low enough to gain volume, thus preventing other competitors from going down the learning curve and hoping to emerge as low-cost producers with dominant market shares.

Product cost, then, is not a simple “hard” number. How cost is calculated for pricing purposes is a matter of managerial judgment. It may be construed as full cost or as out-of-pocket cost. It may be the cost levels being experienced or experience curve estimates of future costs. The interpretation of cost factors for pricing will depend greatly on product/market objectives (a topic also treated below).

Customer Value

Some business managers set prices simply by adding a percentage over costs to provide an acceptable profit. That approach has two advantages. Price is simple to calculate and if a firm is a low-cost producer, relative to competitors, so-called “cost-plus” pricing may seem to provide some protection from competitive attack.

The trade-off for simplicity and security may be lost profits. In theory the amount of profit that is sacrificed is the difference between what customers actually paid and what they would have been willing to pay.

Compared with cost-plus pricing, pricing according to the value of the product to the customer is more difficult and conjectural. How does one determine what the value of the product is in the customer’s mind?

First, it is useful to distinguish between perceived value and potential value. Perceived value is what the buyer now recognizes. Potential value is what the buyer can be educated to see in the product. That is a task of marketing. It may be accomplished through advertising, personal selling, and getting the buyer to try the product.

Second, product value may be perceived differently by different customer groups or market segments. Different segments may place different values on the several elements that make up the set of product attributes. (*Product*, here, is to be taken in the broadest sense to include the product or service itself, its brand image, its availability, and the service that the seller provides.) A business person traveling by air will place little or no value on the fact that the hotel he or she stays in offers free parking facilities. But that may be an important consideration for a family group vacationing by car. A large firm may place little value on the technical service a supplier offers if it (the large firm) has comparable or superior technical resources of its own. But a small company may be highly dependent on the supplier’s technical services and place high value on them in making purchasing decisions.

A third factor to consider in establishing customer value is the options that a potential buyer may have. Clearly, if the buyer can purchase a product at a lower price from one source than another, the lower price sets the upper bound in the marketplace. A corollary, however, is that for the buyer to have effective options, he or she must have knowledge of them.

The options that different customers may have may vary widely. For example, a manufacturer of shower heads may have a choice between using chrome-plated brass or a high-performance plastic. Given this choice and the exceedingly high cost of brass, the plumbing manufacturer is likely to put a high value on the plastic alternative. If the choice is between a plastic molded part and a steel stamping to be used in making a child's toy, the plastic may not be valued as highly.

Another option the customer has may be not to buy the product at all but to make-do with what one has. Given this choice, the buy-not buy decision may be made by comparing the outcome of one course of action with the other. A family may choose, for example, between keeping its old (gas-guzzler) automobile and suffering the increasing costs of maintenance and fuel or buying a new, fuel-efficient car. An industrial company may choose between keeping an old machine in operation or buying new, more efficient equipment.

These choices are quantifiable. One may calculate the operating savings in either instance and relate this to the cost of buying (or investing in) the new car or new machine tool, as the case may be. The anticipated savings may be expressed as a percentage *return on investment* (ROI). Thus, the amount of realizable savings establishes the value of a product to the customer. Calculated for each of several possible uses for available funds, ROI measures may serve to establish the buyer's purchasing priorities.

The choice between buy-not buy, of course, may not always be easily quantified with reference to expected savings. For example, part of the value of the new car may be psychic, part may be a more comfortable ride. The buyer of a new TV set can hardly establish the clarity of the image it gives and its better color rendition as a savings when comparing it against the performance of the old set. But these are, nevertheless, real and important considerations.

Finally, the price set by the seller is often taken by the potential customer as one measure of the value of the product. It is often interpreted by buyers as the supplier's estimate of the worth of the product it sells. If the seller does not value the product highly, it is not likely that the buyer will. Therefore, pricing a product significantly below what the buyer might pay for its functional equivalent can be self-defeating. The buyer may infer that value is, in fact, connoted by price and choose the higher-priced option.

Value, then, for a given product tends to be a function of (1) the utility of its several attributes to the prospective buyer, (2) the options the buyer has and is aware of (i.e., the offerings of competing suppliers and the option simply of not buying at all), and (3) the extent to which the buyer perceives price itself as a measure of product value.

Price discrimination If the seller truly value-prices, then different prices would be charged for the product to different customer groups. Like the proverbial rug merchant in the Persian bazaar, the seller would negotiate each sale at a price that was mutually acceptable to the seller and the customer. Regarding physical goods, however, that may be an impractical strategy unless the form of the product can be altered so as to persuade customers that the product being sold in one market segment is different from that sold in others. A manufacturer of a heavy-duty cleaner could, for example, use different brand names and different kinds of containers for in-home applications, industrial uses, and the hobby (say, boating) market and charge three different prices for the same chemical formulation. Price discrimination of this sort works only as long as (1) the products sold in one market are not, for all practical purposes, available to buyers in other segments, and (2) buyers in different markets are not aware that they can buy the same product under a different brand name at

different prices. Price differences, however, that reflect variations in the design or formulation of the product to adapt it to the needs of different market segments are more sustainable.

In the marketing of services, however, price discrimination among classes of buyers is widely practiced. An airline may charge one fare for adults, another for children, another for spouses, another for standby passengers and a different rate for service personnel. These rates may vary between weekdays and weekends, summer and winter, day and evening. Hotels have similar pricing strategies. Another example is computer-based information services, which may have one rate for a peak-hour usage and another for off-peak; one for libraries and another for business firms.

The reason why price discrimination is possible for services is that they have high time utility. A room in the Waldorf-Astoria unoccupied on a Sunday night cannot be saved and sold Monday, or shipped to Chicago, put in inventory, and sold during the busy fall months. The same would be true of an airline seat on any given flight, a 30-second spot TV commercial, a seat at a Red Sox-Yankees baseball game. If each of these services is not sold at the time it is available, it is lost forever.

Price sensitivity A relevant consideration in thinking about price as an expression of product value is how sensitive is the buyer to price. Price sensitivity will vary considerably among purchasers and, for the same purchaser, it will vary from one time and one set of circumstances to another. Buyers who can pass on the cost of the purchase are less sensitive to price than those who cannot. A business person who can charge expenses to his or her company is less price sensitive to travel costs than a family vacationer.

Price sensitivity also relates to the performance standards by which the purchaser is measured. An engineer buying process equipment for a new chemical plant will be less sensitive to price than to assuring that the equipment is delivered on time, does not have start-up problems, and gives trouble-free service. What the equipment costs is less important to the engineer's performance measures than how it performs. Consequently, he or she may be willing to pay a price premium to a supplier who has a reputation for high-quality standards and exceptional service.

Viewed another way, performance measures effectively establish the relative worth of different product attributes for the manager who has to make the decision and be judged for it. In industrial buying situations, therefore, purchasing managers are characteristically more price sensitive than engineers because each is measured along different dimensions.

A factor in price sensitivity is the uncertainty that attends switching from one supplier to a lower-priced source. Modest price differences are often insufficient to overcome the purchaser's uncertainties about an untried supplier's product quality, reliability, and service. Moreover, there is often the concern of being locked in to some new sourcing arrangement and vulnerable to subsequent price increases. Thus, price differences between the "in" supplier and the "outs" may have to be significant to get the buyer to switch from a known and comfortable buyer-seller relationship to a relatively unknown source of supply.

Competition

Competitive market price levels usually impose a tight discipline on value pricing. They reflect not so much what the product is worth to the customer in some absolute sense as the availability of supply relative to demand. The greater the supply relative to demand, the lower the price.

For so-called commodities (i.e., virtually undifferentiated products), all competitors generally charge identical prices. If one goes above the market price, sales will drop off sharply; if one goes below, all others are likely to follow or risk significant reductions in market share.

How much any individual firm is constrained by competitors' prices, therefore, depends largely on how differentiated its product is. A product that is set apart from other market offerings by its functional design, appearance, brand image, and the supplier's reputation for service and availability in ways that have value to customers can command a price premium. IBM, for example, has typically charged prices for computers over what its competitors have charged for comparable equipment. Its customers are persuaded that IBM offers superior systems design and field service and is in the forefront of computer technology.

There are circumstances, however, in which firms will price over competitive levels even though the price differences are not really justified by superior product quality and service. A company may consciously elect, for example, not to meet competitive prices in a strategy of "milking" the business, that is, yielding market share and gradually withdrawing from the market. It may continue to sell profitably for some time to its loyal customers, in the meantime gradually cutting back on selling and promotional expenses until it eventually phases out of the market.

Some companies may choose not to price competitively because to do so would mean selling below cost. These so-called marginal firms eventually go out of business.

Some large companies may not elect to meet the low price of a smaller competitor because to do so might mean giving up unit profits on a large sales base. It may be less costly in the short run to hold prices and give up some small percentage of market share. In the long run, the smaller competitor encroaches increasingly on the market positions of its major competitors until it becomes, itself, a major factor.

Under shortage conditions some firms may price opportunistically above prevailing market levels, knowing that demand far exceeds available supply and that some buyers will pay the high price.

Finally, some firms may unknowingly be underpriced by competitors on some of their products. These products may be part of a broad line and the reporting system may not allow for monitoring the sales-profit performance of each item on the list. Thus the company may be losing sales and market position because of price and never realize, until too late, that the business has gone to more aggressive competitors.

Generally, however, pricing strategies must inevitably be shaped with regard for present and future competition. In this respect there is significant pricing interdependency among firms in an industry with each being heavily influenced by the others' strategies and tactics. Some firms follow price trends; others, the larger ones, seek to lead them. Accordingly, in contemplating price changes the marketing manager will often seek to anticipate competitive responses. "If we do this, what will they do?" Price leaders, moreover, will plan their moves in ways designed to elicit certain anticipated responses from competitors.

Price leadership What is a price leader and what are the qualifications for filling that role? A price leader is the dominant influence on market prices for a class of products. By initiating price increases or decreases the leader may give direction to market price patterns. One may lead without moving at all. If, for example, others seek to raise prices and the leader doesn't follow, the upward move falters.

This is not to say that other firms do not influence price levels. On the contrary, it is often the smaller firms in an industry that undermine the prevailing price level through shading prices in a series of transactions.

Price leaders can enforce a discipline on market prices only to a limited extent; the general price level tends strongly to reflect supply-demand forces. The leader does act, however, as the dominant, or reference-point firm among competing sellers of a product line.

Price leadership is a phenomenon observed primarily in oligopolistic industries. The leader generally has the largest (or a large) market share, is a low-cost producer, has strong distribution, and is often recognized as being in the technical forefront of product and process development. In particular, the leader earns and holds its position through having initiated price moves that are followed by competitors and accepted by customers. False moves tend to destroy the leader's credibility among its competitors.

Other factors contributing to the industry acknowledgment of a price leader is a tacit recognition that the dominant firm's low-cost position and resulting high margins give it the resources for funding new product development, market development, and extensive promotional programs. It is recognized, as well, for having superior financial resources and hence superior capacity for withstanding economic reverses. Finally, there is the implicit assumption that having a large market share, the leader will act in the overall industry's best interest because it is in its best interests to do so.

The effective exercise of a leadership role is contingent on several factors. First, it is essential that the leader have a superior market information system as the basis for understanding what's going on in the market and reacting in a timely way. Another condition of effective leadership is a clear sense of strategy. A third is long-term (more than one year) measures of managerial performance. Managers in leadership firms may be required to act in the short term in ways that adversely affect market share and/or profitability. A strategic plan in which price is a key factor may take two to three years to see through to a profitable outcome.

Finally, price leaders must *want* to lead. They must be willing to take that responsibility and to act responsibly. They must have a broad concern for the health of their industries that goes beyond selfish interests.

What are some of the characteristic pricing strategies of leader firms? In oligopolistic industries, the leader will tend to behave in a way that preserves short-run, market-share stability. To seek aggressively an increase in market share is to invite sharp competitive price reactions and a resultant reduction in overall industry profit levels.

In addition, industry leaders may even yield market share to smaller, weaker competitors to forestall their price cutting in an effort to preserve sales volume. There is a general tendency for individual firms in oligopolistic industries to seek a share of market that approximates its share of industry capacity. Because of this tendency, market prices in conditions of general underutilization of capacity tend to be set by the firm that has the least favorable ratio of market share to share of industry capacity. It seeks to redress the balance by reducing prices to improve volume—and all competitors are forced to follow. A way for the leader to preserve price levels generally in economic downturns, then, may be to yield market share, if possible, to competitors whose market share/plant capacity ratio is such as to lead them to cut prices. The leader may then take share back on the economic upturn by lagging slightly behind the rising price trend.¹

Because price leaders are often technical leaders, their strategies also take into account the relationship between the prices of new products and present offerings. New products, superior in performance to the ones they will replace, are usually priced at a premium over the old ones, commensurate with the increased value to customers. This tends to preserve price levels and margins in the old lines and give competitors less incentive to cut prices to preserve volume. The strategy is a reinforcing one; the higher margins on the new products provide overhead funding to support continued product development and a strengthened leadership position.

¹ For a detailed study of competitive behavior in the heavy electrical equipment industry see Ralph G. M. Sultan, *Pricing in the Electrical Oligopoly, Volume 1, Competition or Collusion* (Division of Research, Harvard Business School, Boston, Massachusetts 02163, 1974).

Government Influences

In addition to competitors, government agencies also strongly influence prices in many ways. In regulated industries such as utilities, airlines, and railroads, state and federal commissions have the authority to approve or reject price changes. Their concern is usually to protect consumer interests but to allow the regulated company enough profit to expand capacity as demand grows and a reasonable return on its investment.

The so-called nonregulated industries are subjected, as well, to government influence on their pricing actions. Agencies such as the President's Council on Wage-Price Stability (COWPS) set price guidelines based usually on inflation factors, monitor announced price changes, and often act vigorously to get firms to roll back increases that exceed the guidelines. In 1978, for example, COWPS successfully took action against Sears, Roebuck to get this large retailer to roll back the prices published in its fall 1978 general catalog. Given a long history of periodic government efforts to hold down price increases, there is evidence that business managements are loath to reduce prices in an economic downturn for fear of lowering the baseline against which subsequent price increases might be measured.

Other sources of government influence on prices, of course, are such legislative actions as the Sherman Act and the Robinson-Patman Act. In general the Sherman Antitrust Act seeks to preserve competition by proscribing predatory pricing (i.e., pricing that has the intent or effect of driving firms out of business). The Robinson-Patman Act makes it illegal to sell the same product at different prices to different customers who are, themselves, in competition with each other. Granting one customer a lower price than is being charged to its competitors, however, is permissible under either of two conditions: (1) if a low price to one customer meets the equally low price being offered to that firm by a competitor, or (2) if the reduced price passes on a cost savings that may result from lowered manufacturing, transportation, or marketing costs associated with the sale.²

A Sense of Fairness

In the absence of government restraints in periods of shortages, a natural tendency for those who control the supply of a product is to raise prices to reflect the supply / demand imbalance. Public pressures, however, such as those brought to bear by citizen groups in the gasoline shortages of 1978 and 1979, may often be powerful restraints. But business managers may often exercise such restraint voluntarily because of their sense of what's fair. Fairness has a high value in the relationship between buyer and seller. If a buyer feels that he or she was gouged when market conditions gave the seller the upper hand, the seller may damage its longer-term customer relationships and erode its market base.

Pricing Objectives

Pricing is a key element in strategy, and to make strategic pricing decisions it is important to know what objectives are being served. The range of possibilities is quite large. One may seek to gain market share or to yield it opportunistically with little concern for future consequences (i.e., "milk" the business). A seller may price to discourage some competitors and forestall others from entering the market. Or a seller may hold prices high (a "price umbrella") to avoid driving out less efficient competitors and possibly risk being taken to court on anti-trust charges. Last, a seller may seek to maximize short-run profits or to "buy in" with the hopes of making profits in the long run.

² See the "Note on Legal Issues in the Pricing Process," (No. 578-205) by John F. Cady for an in-depth treatment of the regulation of pricing practices under the Sherman Act and the Robinson-Patman Act.

One may price low to meet a competitive attack, or to take an order that offers an opportunity to gain a new customer, or to acquire experience in designing and making certain products. A seller might also price a product to break even or even to sustain losses on some products in the line to be able to offer customers a full line. Alternatively, one might price high on some new product to minimize its impact on the sales of old products in the line (avoid cannibalization). Whatever the objective, it is important that it be determined explicitly. Otherwise pricing decisions become aimless responses to the moves of others.

Skimming and penetration pricing Market entry situations and the development of new markets raise particular questions of pricing objectives and strategies. Marketers often face the choice in the early stages of the product life cycle of pricing high (skimming) to maximize short-term unit contribution or pricing low (penetration) to maximize unit volume and preempt competition.

A skimming strategy has the advantage of permitting an early-stage focus on those customers for whom the product has the greatest value and who will pay the most. Then, as the price comes down, new market segments open up, theoretically, in order of declining product value to different customer sets. A strategy of reducing price gradually to broaden the potential market, in principle, maximizes total profits. It may also serve the long-run development of the market by establishing a prestige image for the product in the introductory phases. Polaroid, for example, first introduced the Land camera in 1948 at over \$200, gradually bringing out new models until in 1965 it had one model, the Swinger, priced at under \$20.

A skimming strategy on the part of the innovator, however, may be an open invitation to competitors to enter the market and exploit the new product at lower price levels. This is typical, for example, in high-fashion merchandise. New styles introduced to a selected clientele in Paris at prices only the wealthy can afford find their way quickly to New York shops at a fraction of the original price and then to the bargain basements of the big department stores. The originator may be content to take the cream off the top and leave the mass market to copiers—or may have no choice in the matter.

Penetration pricing, in contrast, incurs high risks and offers potentially high rewards. There are several conditions that must obtain if the gamble is to pay off. First, the product should be free of any possible defects, or else the seller runs the risk of generating high demand with the announced low prices and then facing extensive field service problems and/or product recalls.

Second, potential customers should be able to adopt the product quickly without having to test it for long periods, or else competitors will have time to ready their own market programs. Third, sufficient plant capacity should be in place and distribution channels, as well, to fill market demand quickly. Success in penetration pricing hinges critically on timing and on taking away from competitors the opportunity to react.

Examples of successful penetration pricing strategies are Bic's taking over the lion's share of the ballpoint pen market and Texas Instrument's rise to dominance in the hand-held calculator market. Both aggressively cut prices and emerged as market leaders. According to industry sources, TI "priced ahead of the experience curve" in the early stages of market development, willingly sustaining losses, and then eventually becoming profitable as many smaller competitors dropped out of the business.

These examples suggest another condition that should be satisfied for penetration pricing to make sense: there should be vast potential demand to be tapped at lower price levels. A penetration price strategy is not likely to be effective in mature, slow-growth industries dominated by entrenched competitors. Their capacity for fighting off the invader by reducing their prices to variable cost levels would hardly "make the game worth the candle" for the new entrant. Penetration pricing is also unlikely to be effective in new fields if the product, even though priced close to cost, is not affordable for most potential purchasers.

Demand elasticity A consideration of penetration pricing brings up the matter of demand elasticity as it relates to pricing strategy. In economic theory, the lower the price the greater the demand will be for the product. In developing pricing strategy, however, some modifications of the theory must be recognized. The demand for many goods, particularly in the industrial sector, is derived demand. The demand for truck motors depends directly on the demand for trucks, for example. Price reductions on this component are not likely to increase truck motor industry volume.

Furthermore, price reductions in the early stages of market development for a product that customers must be educated to use are not likely to stimulate sales. It will take more than attractively low prices at this stage to overcome inherent market caution. A new plastic material might serve as an example.

Price elasticity for any particular product most likely reflects the effect of users switching from one product or service to another as significant price differences occur among substitutable products. As gasoline prices rise relative to the cost of public transportation, more people will travel by air, train, or bus instead of by car. As ballpoint pen prices drop, sales of pencils decline.

In setting pricing objectives, therefore, it is important to ask a range of questions: What is doable? Where will sales volume come from? How will my competitors react? What will be the impact of the pricing strategy for one product on other products in the line? How will potential customers react? But the most important question of all is *What are we trying to do?*

The Pricing Process

In economic theory, prices are set at the intersections of supply and demand curves. But for the individual firm this notion offers little help. Supply / demand analysis may be broadly relevant for an entire class of products such as wheat or nylon fiber or beef or television sets. But unless the product of one supplier is unique and has no direct competition (i.e., as in a monopoly), the single firm's price is necessarily a function of market price levels for that class of products, adjusted for any differentiation between that firm's products and those of reference-point competitors—as customers perceive the difference.

The objectives of the pricing process, then, are first to establish the market value over competitive prices of the differentiated product. A second purpose may be to probe and adjust for any shifts in supply / demand conditions and to test the willingness of competitors to follow. Finally, a purpose of the pricing process is to adjust price to reflect cost factor changes and to prevent cost escalation from eroding margins.

The pricing process is a reiterative exercise. Like game-playing, each decision is usefully perceived as one move in a series of plays. Pricing strategies can be carried out effectively if:

- Pricing decisions are based on extensive current market information.
- Competitive and customer responses to price changes are carefully monitored and recorded.
- Pricing decision making is centralized within a business and not delegated to field representatives. (Centralized control is critical for formulating and implementing price strategies.)
- Pricing moves are planned responses to market conditions and competitive behavior.

Announcing price changes An essential part of the process is the implementation of price decisions. It is not enough to decide what the price should be; it is important as well to prepare customers and competitors to accept it. Consumers as well as industrial procurement managers must understand for themselves and often explain within their organizations (the family unit or the business) the reasons for significant price changes. Competitors then have to make decisions about whether they will follow.

The great bulk of price increases are publicly explained as adjustments for cost factor escalations; others are explained as reflecting product improvements. To accept price increases, customers must be persuaded, if possible, that they are fair and are justified by changes in cost economics or product value. How the seller announces the price increase is therefore a critical part of the pricing process.

Pricing Modes

Prices are expressed in various forms for both consumer and industrial products. It is important for marketers to know the range of practices. In particular, it is useful to understand that a way of stating price often has strategic implications for either the seller or buyer or both. Perhaps the simplest way to proceed is to look first at consumer goods pricing conventions and then turn to examine the industrial side.

Pricing Consumer Products

Consumer goods sold through wholesalers and retailers are priced to reflect the margins accruing to these resellers. Margins, unless otherwise stated, are expressed as percentages off the list price to the consumer. An item selling for \$1, for example, might be priced to provide the retailer a 40% margin and the wholesaler a 10% margin. These margins are taken successively. Thus,

Manufacturer's suggested retail price (MSRP)	\$1.00
Less: retail gross margin (RGM) at 40% of \$1.00	<u>.40</u>
Wholesaler's price to retailer	.60
Less: wholesale margin at 10% of \$.60	<u>.06</u>
Manufacturer's selling price (MSP)	\$.54

In the above example the manufacturer sells to the wholesaler at 40% + 10% (a total of 46%) off MSRP or \$.54, and the wholesaler sells to the retailer at 40% off list price or \$.60. Thus, the wholesale margin shown is \$.06 (10% of \$.60).

The fact that the manufacturer may suggest a retail price, however, in no way binds the retailer to sell at that price. But if the retailer sells below that price, any difference must come out of the margin the seller would receive if it had, in fact, sold at the MSRP.

Some manufacturers selling through wholesalers to retailers may simultaneously sell directly to large retail accounts such as supermarket chains. In this situation the manufacturer has a choice. It may sell to wholesalers and to large retail accounts at the same price on the theory that the large retailer is actually performing both wholesale and retail functions. Alternatively, the manufacturer may protect its wholesalers by selling to retail accounts at the same price the wholesaler would charge to retailers. Usually the choice depends on the importance of wholesalers in the manufacturer's overall marketing strategy. If they are important and account for a high percentage of the manufacturer's sales, the manufacturer will not want to risk its good relations with its wholesalers by undercutting them in their markets.

The manufacturer's selling price often varies with the quantity purchased. The larger the order, the lower the unit price. Here, for example, is one manufacturer's discount schedule for furniture polish sold in cases of 24 one-quart cans.

Cases to Wholesalers

1–99	$47\frac{1}{2}$	%
100–400	$47\frac{1}{2}$	+ 2.5%
401–900	$47\frac{1}{2}$	+ 4.0%
901–above	$47\frac{1}{2}$	+ 6.5%

Cases to Retailers

1–5	30	%
6–above	30	+ 10%

In addition to volume discounts, other percentage discounts may be taken off the manufacturer's prices to wholesalers and retailers. For example, the manufacturer may give a cooperative advertising allowance of 5% of sales as an inducement to resellers to advertise its products. Thus, the manufacturer may offer to reimburse the reseller for half the cost of local advertising up to a maximum of 5% of the reseller's purchases.

Often manufacturers and/or wholesalers also offer their respective customers percentage discounts off the purchase price to encourage prompt payment. The bill, for example, might include a statement such as "terms—2%, 10 days, net 30 days." That means that if the buyer pays within a 10-day period, the buyer may deduct 2% from the total bill and that the full amount is due in 30 days.

Some manufacturers sell on consignment to their customers. In consignment selling, title to the goods remains with the manufacturer until sold by the reseller and, at that time, payment is due. Consignment selling relieves the reseller of carrying the investment in inventory and puts that burden back on the manufacturer.

An important consideration in consignment selling is that the manufacturer may legally specify the retail selling price since it owns the goods until they are sold to the ultimate purchaser.³ The reseller, in fact, acts as the manufacturer's agent in consignment selling. Thus, manufacturers may offer consignment arrangements to resellers for one or both of two reasons: to persuade the reseller to carry the goods and to control the retail selling price.

Pricing Industrial Products

Many of the conventions used in consumer goods pricing are also practiced in industrial marketing. Industrial goods pricing may also include reseller margins and provide discounts for purchasing in volume, for paying bills promptly, and sometimes for cooperative advertising.

An industrial counterpart of the furniture polish pricing schedule shown above would be the following one for grinding wheels, in which different prices are quoted for different purchase quantities and two different classes of purchasers, distributors and end users. Notice that prices shown are expressed as percentages of a list price, not shown here.

³ The legality of using consignment arrangements as a way of fixing resale prices has been tested in the courts over a long period of time and is still unclear.

User-Customer and Distributor Multipliers

Quantity Group	No. of Units	User-Customer Multipliers	Distributor Multipliers
<A	1–9	1.027	.822
A	10	.689	.551
B	20	.585	.476
C	50	.503	.418
D	100	.447	.393
E	250	.394	.355
F	750	.379	.341
G	1,250	.365	.328

Source: Norton Company

Another pricing convention used in pricing both industrial goods and consumer goods has to do with freight costs from the point of shipment to destination. Typically the price quotation specifies whether these charges will be paid by the seller or the buyer. If the price is specified as *f.o.b.* (freight on board) *plant*, the buyer pays the freight charges. If *f.o.b. destination* is stipulated, transportation costs will be paid by the seller.

In industrial goods marketing, freight charges and who pays them may have considerable competitive significance. For products such as cement and steel, transportation costs are relatively high compared with the value of the product itself. High shipping costs naturally serve to limit the area within which a manufacturing plant can effectively compete. To extend their markets geographically, producers of products having high weight or bulk relative to value may use *basing-point pricing* (also called freight equalization). In this system, the transportation cost component of a quoted price is the freight to the buyer's plant from the nearest source of supply. That source of supply often is the plant of a competitor. Although basing-point pricing is a way of competing for distant customers, it also means that by absorbing freight the seller nets less revenue per unit from these accounts than from ones closer to its plant.

Much industrial buying is done by competitive bidding. One method is the sealed bid in which all bids must be submitted before a given time on a specified date. The bids are opened publicly at the announced time and the lowest qualified bid is awarded the contract. *Qualified* means that the seller's bid meets the specifications issued by the buyer in the RFQ (request for quotation).

A negotiated bidding procedure differs from sealed bidding in two respects. Bids are not necessarily made public, and the buyers may continue to negotiate with the one or two lowest bidders rather than accept the lowest initial offer.

Prices specified in purchase contracts may be fixed-fee or cost-plus or target-incentive. In contracts negotiated as cost-plus arrangements, the buyer agrees to pay the seller certain costs of filling the contract plus an amount to cover fixed overheads and profit. Typically, the costs that are paid directly include labor and material. Overheads-plus-profit may be charged on a per-labor-hour basis. In addition, any capital investments needed to do the work may be amortized as part of the unit price until they are completely written off.

In a cost-plus contract the buyer usually negotiates audit privileges with the seller. That is, the buyer has the right to inspect the contractor's cost records.

Some contracts may provide for cost-factor escalation. These provisions allow for price adjustments as certain specified costs (e.g., labor and materials) increase. Contracts specifying cost escalation must stipulate what particular price indices will be used to determine the amounts of price adjustments. Often these indices are ones published by government agencies such as the Bureau of

Labor Statistics in the Department of Commerce or those issued by trade associations, based on market price time series.

Finally, some contractual arrangements are intended to provide cost efficiency incentives to the contractor. These are target-incentive contracts. They specify a target cost, a ceiling price, and a profit-sharing formula. If the contractor beats the target cost figure, the contractor splits the savings in a prescribed ratio with the customer. If the target cost is exceeded, the contractor pays part of the difference and the customer pays the rest. The customer is obligated, however, to pay no more than the ceiling price.

Such a ranging review of pricing conventions cannot be all-inclusive. The variety of arrangements is limited only by the ingenuity of buyers and sellers negotiating with each other day-by-day in the marketplace. They reflect the strategic objectives of the latter shaped in recognition of the needs and concerns of the former. What has been covered here, however, are among the most widely used and recognized practices.